

Skills Summary

Graphing Proportional Relationships

Use this reference while completing your worksheet or when studying.

Reading the Unit Rate from a Graph

Rule: a proportional relationship graphs as a straight line through the origin $(0, 0)$. Its **unit rate** is $k = \frac{y}{x}$ at any point on the line — and it is exactly the y -value sitting above $x = 1$.

Example: A line goes through $(0, 0)$ and $(4, 14)$. Find the unit rate.

Step 1. Any point on the line gives the rate — divide y by x at the clear point $(4, 14)$.

Step 2. $k = \frac{14}{4} = 3.5$
 $\Rightarrow$ **3.5**

Try it: A line goes through $(0, 0)$ and $(6, 15)$. Find the unit rate.

check: 2.5

Writing the Equation from a Graph

Rule: every proportional relationship has the equation $y = kx$, where k is the unit rate read from the graph. Once you know k , the equation predicts y for *any* x .

Example: A proportional graph passes through $(5, 30)$. Write its equation.

Step 1. Find k first, then drop it into $y = kx$.

Step 2. $k = \frac{30}{5} = 6$
 $\Rightarrow$ **$y = 6x$**

Try it: A proportional graph passes through $(8, 20)$. Write its equation.

check: 2.5

Graphing from a Table

Steps: (1) check the table is proportional: every row gives the same ratio $\frac{y}{x}$; (2) plot each (x, y) pair; (3) the points line up on a straight line through the origin — draw it. If the ratios differ, the points will *not* line up, and the relationship is not proportional.

Example: A table gives $(2, 7)$ and $(4, 14)$. Find the unit rate before plotting.

Step 1. Check both rows give one rate — that is what makes the line straight.

Step 2. $\frac{7}{2} = 3.5$ and $\frac{14}{4} = 3.5$ — same rate, so the points lie on the line $y = 3.5x$.
 $\Rightarrow$ **3.5**

Try it: A table gives $(3, 12)$ and $(6, 24)$. Find the unit rate.

✓ check